Development of an All-Territory-Compatible Drone for Autonomous Exploration

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1 Project Goal

The goal of this project is to develop an All-Territory-Compatible drone capable of handling any situation.

2 Introduction

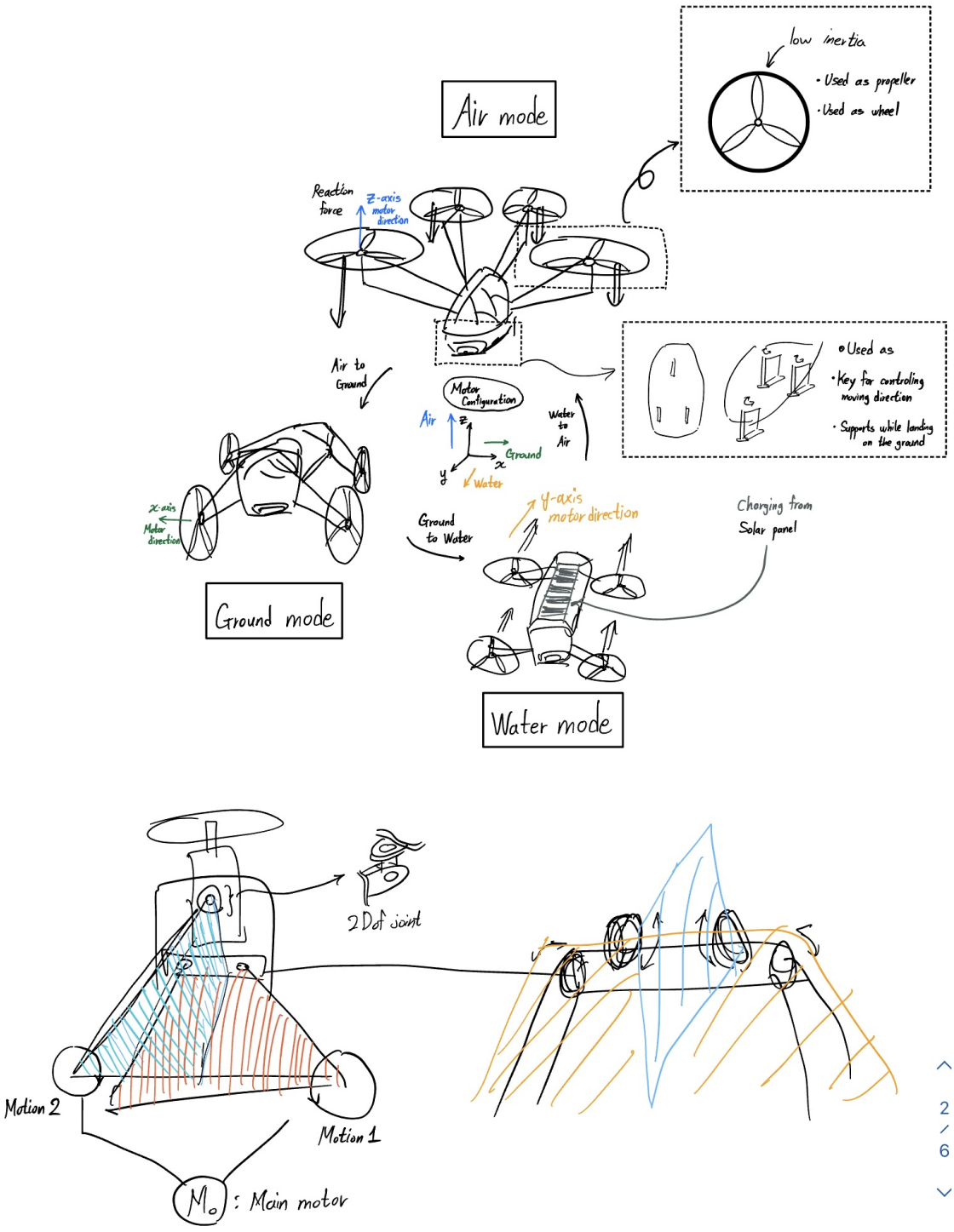
With the advancement of artificial intelligence (AI), the development of unmanned systems requiring less human intervention has significantly increased. In particular, exploration tasks that require significant effort and carry a high level of risk have seen a growing reliance on robots, including exploration probes and reconnaissance robots. Furthermore, drones are essential for mapping areas, but their high cost necessitates automatic retrieval and operation. Reducing human intervention in situations where large amounts of information need to be collected and filtered is possible if drones can automatically perform retrieval and tasks.

3 Problem Statement

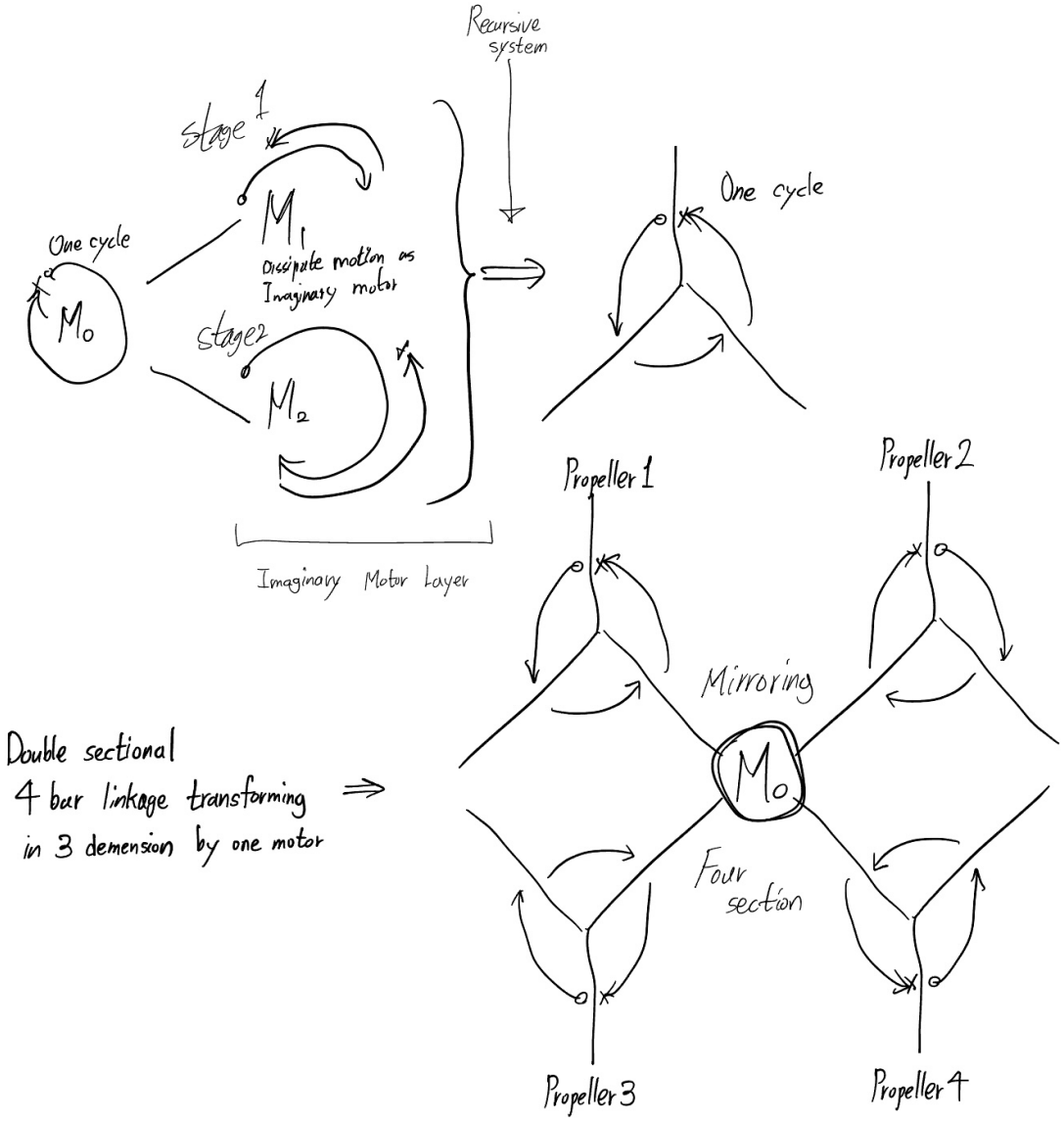
1. Energy efficiency during flight time is lower than ground-based robots. 2. A drone capable of switching to multiple terrains to enable an automated reconnaissance system is necessary. 3. Light weight is the most critical problem for drones, where even a weight difference of 1g can significantly affect flight efficiency.

4 Mechanism

The double 4-bar loop mechanism applied as a cross-section lies in its capability of transforming into any desired vector (x, y, z). Each vector norm(x-axis, y-axis, z-axis) represent configuration of which environment is air, ground, and water. The propeller which is used to lift in the air mode and move forward on the water is also used as wheel at the ground mode. The mechanism is focused on the transition of configuration[BM19].



Figur 1: The transition of the propeller configuration



Figur 2: The coupled motion of four propeller by one motor

The secondary mechanism which can be used is separation of the movement. The first motion(Air[z-axis] to ground[x-axis]) is likened to folding motion of the aircraft. After the first motion arrived the end position the catching motion which rotating[x-y plane] the inner shaft will proceed.

5 Advantages and Novelty

1. The mechanism was designed to reduce weight for both transition and using the same motor for the propeller and wheel. 2. The use of one motor allows for the vector of all four propeller motors to transition in arbitrary 3D Euclidean space, which is a unique advantage. 3. Two sets of four-bar linkage can represent all vectors within 3D space.

Referenser

[BM19] Nathan Bucki och Mark W Mueller. ”Design and control of a passively morphing quadcopter”. I: *2019 International Conference on Robotics and Automation (ICRA)*. IEEE. 2019, s. 9116–9122.